Canada Arm 3 Kinematic Simulations — Nature Preprint

Abstract

This preprint presents a simulation and analysis of the Canada Arm 3 robotic manipulator using a unified web-based interface. Both forward and inverse kinematics are demonstrated, with workspace visualization and finite math calculations for stability and control.

1. Introduction

Canada Arm 3 is a 6-DOF robotic arm designed for space applications. Accurate kinematic modeling is essential for trajectory planning, workspace analysis, and mission safety.

2. Kinematic Model

Joint Limits:

Joint 1 (Base): -170° to +170°

Joint 2 (Shoulder): -90° to +135°

Joint 3 (Elbow): -135° to +135°

Joint 4 (Wrist 1): -180° to +180°

Joint 5 (Wrist 2): -120° to +120°

Joint 6 (Wrist 3): -360° to +360°

Sample Pose: [30, 45, -20, 90, 0, 180]

End Effector Target: [1.2, 0.5, 2.0] (meters)

3. Forward Kinematics

Given the sample joint angles, the forward kinematics yields:

- End Effector Position: (see UI for computed values)
- Workspace: Visualized as a 2D projection in the UI.

4. Inverse Kinematics

Given the target position, the inverse kinematics yields:

- Joint Angles Solution: (see UI for computed values)

5. Finite Math Calculations

State-space stability analysis is performed for the kinematic system.

Eigenvalues and transition matrices are computed for the linearized model.

6. Results

Both FK and IK are demonstrated interactively in the web UI.

Workspace and solution logs are visualized.

Stability and eigenvalue results are available in the Finite Math tab.

7. Conclusion

This unified simulation demonstrates the Canada Arm 3's kinematic capabilities and provides a foundation for further research in space robotics and control.

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